Localized Bump Response Threshold Release for HAOS-IIP

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Abstract

This paper freezes release `53.2` of HAOS-IIP as a localized-bump response threshold result on the scalar carrier. The previous `52.5` smooth-inhomogeneity release left localized bump response open, with maximum refinement drift `0.5124`. Release `53.2` keeps the same scalar carrier and same localized bump profile, but excludes the earliest source-core transient layer using fit window $[0.010, 0.026]$. Under that bounded readout, all ten weak localized bump cases pass for amplitudes $\eta = 0.01, 0.02, 0.03, 0.04, 0.05$, with maximum refinement drift `0.1095`, maximum median relative error `0.0943`, maximum p90 error `0.2984`, and maximum shell-κ CV `0.1248`. Stronger localized bumps remain open: the stress window $\eta = 0.075, 0.10, 0.15$ reaches maximum drift `0.3578`. The correct conclusion is therefore thresholded: weak localized bump excitations close as a bounded metric-tracking transport response, while stronger localized excitations remain outside the same closure.

Contents

1 Scope

Release `53.2` is a narrow follow-on to the physics bridge and scalar-carrier transport stack.

It does not modify HAOS core, add new dynamics, claim particle ontology, or introduce stress-energy coupling.

It only asks:

> does the localized bump response recover if the source-core transient layer is excluded from the constitutive fit?

2 Construction

The test uses:

- the same scalar kernel-graph carrier;
- the same localized bump shape with $\sigma = 0.18$;
- refinements $n = 13, 15$;
- delayed fit window $[0.010, 0.026]$;
- the same current-closure thresholds used by the `52.5` inhomogeneity release.

Two amplitude regimes are separated:

- weak: $\eta = 0.01, 0.02, 0.03, 0.04, 0.05$;
- stress: $\eta = 0.075, 0.10, 0.15$.

3 Frozen Result

Regime	Cases pass	Max drift	Max median err.	Max p90 err.	Max shell CV
weak	10/10	0.1095	0.0943	0.2984	0.1248
stress	0/6	0.3578	0.2784	0.5287	0.3406

Table 1: Weak localized bump response passes; stronger localized bumps remain open.

The weakest passing weak case is:

- `weak_bump|n=13|eta=0.050`;
- $\kappa/D_{\text{eff}} = 0.9699$;
- median relative error `0.0943`;
- p90 relative error `0.2984`;
- shell-κ CV `0.1248`;
- $|\text{metric tracking corr}| = 0.9752$.

The hardest stress case is:

- `stress_bump|n=13|eta=0.150`;
- median relative error `0.2784`;
- p90 relative error `0.5287`;
- shell-κ CV `0.3406`.

4 Interpretation

The old broad statement was:

> localized bump response remains open.

The updated statement is:

> weak localized bump excitations close as a bounded metric-tracking transport response after the earliest source-core transient layer is excluded, while stronger localized bumps remain outside the same closure.

This is closer to a proto-particle diagnostic than the previous result, but it is not a particle model. The result only establishes a weak localized-excitation proxy and an amplitude boundary on the scalar carrier.

5 Artifacts

- script: `numerics/simulations/scalar_kernel_graph_localized_bump_response.py`;
- operator note: `experiments/operators/Scalar_Kernel_Graph_Localized_Bump_Response_v1.md`;
- frozen result: `data/20260423_182539_scalar_kernel_graph_localized_bump_response.json`;
- latest result: `data/scalar_kernel_graph_localized_bump_response_latest.json`;
- summary plot: `plots/20260423_182539_scalar_kernel_graph_localized_bump_response_summary.png`;
- profile plot: `plots/20260423_182539_scalar_kernel_graph_localized_bump_response_profiles.png`.

6 Next Boundary

The next honest step is to refine the transition between $\eta = 0.05$ and $\eta = 0.075$, then test whether the weak localized response survives mild carrier disorder.

7 Conclusion

Release `53.2` turns the localized bump row from an undifferentiated failure into a thresholded result. Weak localized excitations now pass the scalar-carrier transport proxy; stronger localized excitations remain open. That is the correct stopping point: a recovered weak localized response, not a general localized-particle theory.